A continuous parametrisation of phylogenetic trees and application for Bayesian inference

Xavier Didelot[1,2,*]

[1] School of Life Sciences, University of Warwick, Coventry, United Kingdom

[2] Department of Statistics, University of Warwick, Coventry, United Kingdom

[*] Corresponding author. Tel: 0044 (0)2476 572827. Email: xavier.didelot@warwick.ac.uk

Running title: A continuous parametrisation of phylogenetic trees

Keywords: phylogenetic trees; tree space parametrisation; Bayesian phylogenetic inference; Hamiltonian Monte-Carlo

1 Introduction

Phylogenetic trees are usually represented using two components: the topology of the tree which is discrete and the branch lengths of the tree which are continuous. The aim of Bayesian phylogenetic inference is to reconstruct a tree given some genetic data using Bayesian statistics to capture the posterior distribution of the tree (Nascimento et al. 2017). This is usually done using a Markov Chain Monte-Carlo (MCMC) to generate samples from this posterior distribution (Gilks et al. 1996). The MCMC needs to propose updates to both the branch lengths and the topology, with the latter using moves such as nearest-neighbor interchange (NNI) or subtree pruning and regrafting (SPR). However, these moves are not easy to optimise (Höhna and Drummond 2012; Zhang et al. 2020). Conversely, many promising MCMC methods have not been applied in Bayesian phylogenetics because they can only be applied to continuous parameters. This includes in particular some adaptive methods where proposal parameters are learned during sampling (Andrieu and Thoms 2008), as well as gradient-based approaches such as the Langevin and Hamiltonian methods (Roberts and Rosenthal 1998; Neal 2011; Betancourt 2017).

To enable the use of these MCMC methods in Bayesian phylogenetics, a continuous representation of the phylogenetic tree is needed. The Billera-Holmes-Vogtmann (BHV) tree space (Billera et al. 2001) is made of one orthant for each topology, connected between themselves at the points where a branch length degenerates to zero to switch from one topology to another. The BHV space is continuous, but its geometry is complex and intrinsically linked with the nature of phylogenetic trees. Consequently, using the BHV for Bayesian phylogenetic inference is difficult and requires specialised algorithms (Dinh et al. 2017).

Here we propose a new continuous parametrisation of phylogenetic trees, which has the advantage of providing a single unconstrained Euclidean parameter space, with fixed number of dimensions, equal to the number of parameters required to specify the branch lengths of a phylogenetic tree. We show that this new parameter space enables the direct application of generic continuous-parameter MCMC inference methods in a way that was not previously possible.

2 New continuous parametrisation

We consider phylogenetic trees that are binary and rooted, with n labeled leaves. There are $n-1$ internal nodes including the root. Such trees are usually represented as a mixture of discrete and continuous variables (Paradis 2006). The topology of the tree is discrete and there are $(2n-3)!! = (2n-2)!/(2^{n-1}(n-1)!)$ unique possibilities (Felsenstein 1978). The branch lengths are continuous, and there are $2n-2$ branch lengths taking values in $\mathbb{R}$ (one for each leaf and non-root internal node).

Our newly proposed continuous parametrisation is constructed recursively. We consider an arbitrary but fixed order for the n leaves. Let T_i denote the rooted tree formed by the first i leaves and let $L_i = |T_i|$ be the sum of the branch lengths in the tree T_i. The tree T_1 consists only of the first leaf. For each subsequent leaf $i = 2, ..., n$ we introduce two unconstrained parameters, $(a_i, b_i) \in \mathbb{R}^2$. The parameter a_i determines where the leaf i is grafted onto T_{i-1}. The parameter b_i determines the length $\exp(b_i)$ of the new terminal branch leading to leaf i. The parameter a_i represents the length of the path from the root of T_{i-1} to the grafting point of leaf i, following a contour traversal of T_{i-1}. Starting at the root, the traversal follows each branch once away from the root and once back towards it, visiting the first child of each internal node and then the second. If $0 \leq a_i \leq 2L_{i-1}$, the grafting point is within the tree T_{i-1}. If $a_i > 2L_{i-1}$, the grafting point is above the root of T_{i-1} by length $a_i - 2L_{i-1}$. Likewise, if $a_i < 0$ then the grafting point is above the root of T_{i-1} by length $-a_i$. This parametrisation of a_i using a contour traversal is illustrated in Figure 1.

At the graft point, a new node is created with first child equal to what was previously below that point, and second child equal to the new branch of length $\exp(b_i)$ leading to the leaf i. This convention also defines the ordering of the children used in subsequent contour traversals. This process is repeated for all leaves from the second one ($i = 2$) to the last one ($i = n$), each time using the contour traversal on the tree T_{i-1} built previously. We can deduce the following recursive formula for the sum of the branch lengths L_i:

$$L_i = \begin{cases} 0 & \text{if } i = 1 \\ L_{i-1} + \exp(b_i) + \max(0, a_i - 2L_{i-1}) + \max(0, -a_i) & \text{otherwise} \end{cases} \tag{1}$$

Let $\mathcal{C}_n = \mathbb{R}^{2n-2}$ denote the parameter space defined above and let $\mathcal{T}_n$ denote the space of phylogenetic trees with n labelled leaves. An element of $\mathcal{C}_n$ is the vector $C = ((a_i, b_i))_{i=2,...,n}$ and it defines a unique phylogenetic tree $T \in \mathcal{T}_n$. The construction above defines a function $\Phi_n : \mathcal{C}_n \to \mathcal{T}_n$, which is

detailed in Algorithm 1. This function is surjective, meaning that for any phylogenetic tree T there is at least one parametrisation C such that $\Phi_n(C) = T$. To see this, consider any $T \in \mathcal{T}_n$, remove leaf n and its pendant branch, and suppress its resulting degree-two parent. Repeating this operation produces trees with successively fewer leaves. Reversing these steps reconstructs T using the proposed grafting procedure, because every required grafting location is reached by the contour traversal and every terminal branch length $\ell_i > 0$ is represented by $b_i = \log(\ell_i)$. However, the function Φ_n is not injective as there are more than one C such that $\Phi_n(C) = T$. This is a consequence of each grafting point being visited twice by the contour traversal: once as the algorithm goes down the branch and then again as it goes back up the branch. Likewise, a grafting point above the root by length x can be obtained by both $a_i = -x$ and $a_i = 2L_{i-1} + x$. Each grafting point is therefore represented exactly twice if we ignore the possibility of branches of length zero. For a rooted binary tree T with strictly positive branch lengths, there are therefore 2^{n-1} distinct values of C such that $\Phi_n(C) = T$. The duplication of grafting coordinates is needed to achieve continuity: it allows the grafting location to move continuously through the endpoints of the contour and across changes in tree topology, without an explicit jump between discrete topological states.

Algorithm 1 Continuous parametrisation of a rooted phylogenetic tree

Require: Ordered leaves $1, \ldots, n$ and parameters $C = ((a_i, b_i))_{i=2,\ldots,n} \in \mathbb{R}^{2n-2}$
Ensure: A rooted phylogenetic tree T_n

Let T_1 consist of the single leaf 1, which is also its root
$L_1 \leftarrow 0$
for $i = 2, \ldots, n$ **do**
    if $a_i < 0$ **then**
        Extend T_{i-1} above its root by length $-a_i$
        Let x_i be the new root
    else if $a_i \leq 2L_{i-1}$ **then**
        $x_i \leftarrow \Gamma_{T_{i-1}}(a_i)$ where $\Gamma_T : [0, 2L(T)] \to T$ denote the contour traversal of T.
    else
        Extend T_{i-1} above its root by length $a_i - 2L_{i-1}$
        Let x_i be the new root
    end if
    Insert a new internal node at x_i, splitting its branch if necessary
    Make the subtree previously below x_i the first child of this node
    Attach leaf i as its second child by a branch of length $\exp(b_i)$
    Denote the resulting tree by T_i
    $L_i \leftarrow L_{i-1} + \exp(b_i) + \max\{0, -a_i\} + \max\{0, a_i - 2L_{i-1}\}$
end for
return T_n

To show mathematically that the parametrisation is continuous, we need to consider a tree space with a well-defined associated metric. Let $\mathcal{B}_n$ denote the Billera-Holmes-Vogtmann (BHV) tree space (Billera et al. 2001) of rooted phylogenetic trees with n labelled leaves including trees with zero-length internal branches. The Algorithm 1 can be defines a function $\Phi_n : \mathcal{C}_n \to \mathcal{B}_n$ which is continuous. Within each

fixed topology, branch lengths and grafting locations vary continuously with the parameters a_i and b_i. When a grafting location passes through an internal node, the separating internal branch contracts continuously to zero before the adjacent topology is entered. Both topologies therefore meet at the same boundary tree in BHV space. The extension above the root also tends continuously to zero when a_i approaches either endpoint of the contour ($a_i = 0$ and $a_i = 2L_{i-1}$). Applying this argument recursively at each grafting step establishes the continuity of Φ_n.

3 Bayesian inference on the continuous phylogenetic space

We consider a prior distribution on the tree T such that all of the $(2n-3)!!$ topologies are equally likely and the branch lengths ℓ_i follow an Exponential distribution with mean θ, so that:

$$p_T(T) = \frac{1}{(2n-3)!!} \prod_{i=1}^{2n-2} \theta^{-1} \exp\left(-\frac{\ell_i}{\theta}\right) \tag{2}$$

To find the corresponding distribution on C we need to consider the Jacobian of the function $\Phi_n : \mathcal{C}_n \to \mathcal{T}_n$. We consider this Jacobian locally, within a region of $\mathcal{C}_n$ where the resulting topology and the branch containing each grafting point are fixed. Let $\boldsymbol{\ell}^{(i-1)}$ denote the vector of branch lengths in T_{i-1}. If leaf i is grafted in the interior of a branch of length ℓ_j, that branch is replaced by two branches of lengths

$$u_i(\boldsymbol{\ell}^{(i-1)}, a_i) \quad \text{and} \quad \ell_j - u_i(\boldsymbol{\ell}^{(i-1)}, a_i)$$

and a terminal branch of length $\exp(b_i)$ is added. Along any fixed segment and direction of the contour traversal, u_i is affine in a_i, with

$$\left|\frac{\partial u_i}{\partial a_i}\right| = 1.$$

After reordering the branch-length coordinates, the absolute Jacobian determinant of this grafting step is therefore

$$\left|\det \frac{\partial \boldsymbol{\ell}^{(i)}}{\partial(\boldsymbol{\ell}^{(i-1)}, a_i, b_i)}\right| = \left|\frac{\partial u_i}{\partial a_i}\right| \frac{\partial \exp(b_i)}{\partial b_i} = \exp(b_i).$$

The same result holds when the grafting point lies above the root, since the new root branch has derivative -1 or 1 with respect to a_i. Applying recursively gives, for every C in which the grafting

points do not coincide with a node or an endpoint of the contour:

$$\left|\det \frac{\partial \boldsymbol{\ell}(\Phi_n(C))}{\partial C}\right| = \prod_{i=2}^{n} \exp(b_i) = \exp\left(\sum_{i=2}^{n} b_i\right). \tag{3}$$

When accounting for this Jacobian and the multiple representations of each T in the space of C we obtain the probability density function of C:

$$p_C(C) = \frac{1}{2^{n-1}} \exp\left(\sum_{i=2}^{n} b_i\right) \frac{1}{(2n-3)!!} \prod_{i=1}^{2n-2} \theta^{-1} \exp\left(-\frac{\ell_i}{\theta}\right) \tag{4}$$

This density corresponds to the prior distribution of C before any data is observed. After observing data D made of n aligned genetic sequences of length l, the distribution becomes the posterior with density:

$$p_C(C|D) \propto p_C(C) p(D|T = \Phi_n(C)) \tag{5}$$

where the last term is the likelihood of the data given the phylogenetic tree $T = \Phi_n(C)$ obtained using Algorithm 1. This likelihood can be calculated efficiently using the Felsenstein pruning algorithm (Felsenstein 1981) which is used in most phylogenetic inference tools. Many substitution models can be used to calculate this likelihood (Nascimento et al. 2017) and here we use the simplest Jukes-Cantor model (Jukes and Cantor 1969) which avoids having to infer additional parameters.

4 Application of the adaptive Metropolis sampler

To sample from this distribution, we use the adaptive Metropolis sampler (Vihola 2011). This algorithm uses a multivariate normal distribution as the proposal kernel $q(C'|C)$ which is symmetric ($q(C'|C) = q(C|C')$), so that the acceptance probability is simply $\min(1, p_C(C')/p_C(C))$ calculated using Equation 4. The mean of the proposal kernel is the current value C and its covariance matrix is tuned to achieve an acceptance rate of 0.234 (Andrieu and Thoms 2008). We take $n = 5$ as a small example, with $\theta = 2$ as the mean of the distribution of branch lengths. The sampler was run for 2×10^5 iterations, with the first half discarded as burn-in and the remainder sampled every 100 iterations to produce a sample of size 1000. This run took less than a minute on a standard laptop. Figure 2 shows that the trees sampled by the MCMC are distributed as expected under Equation 2. All $(2n-3)!! = 105$ topologies have been sampled, the distribution of the sum of branch lengths and of

a single branch length are as expected, and a majority-rule consensus tree looks completely star-like. The effective sample size was greater than 200 for individual branch lengths as computed using the R package coda (Plummer et al. 2006). The approximate effective sample size of the tree topology was also greater than 200 as computed by the R package rwty (Warren et al. 2017).

5 Application of Hamiltonian methods

We use the dynamic Hamiltonian Monte Carlo (HMC) method implemented in Stan (Carpenter et al. 2017; Betancourt 2017). The target distribution is as defined in Equation 4. The MCMC was run for a total of 2×10^5 iterations, with the first half discarded as burn-in and the remainder sampled every 100 iterations to produce a sample of size 1000. This run took less than a minute on a standard laptop. Figure 3 shows that the trees sampled by the MCMC are distributed as expected under Equation 2. All $(2n-3)!! = 105$ topologies have been sampled, the distribution of the sum of branch lengths and of a single branch length are as expected, and a majority-rule consensus tree looks completely star-like. The effective sample size was greater than 200 for individual branch lengths as computed using the R package coda (Plummer et al. 2006). The approximate effective sample size of the tree topology was also greater than 200 as computed by the R package rwty (Warren et al. 2017).

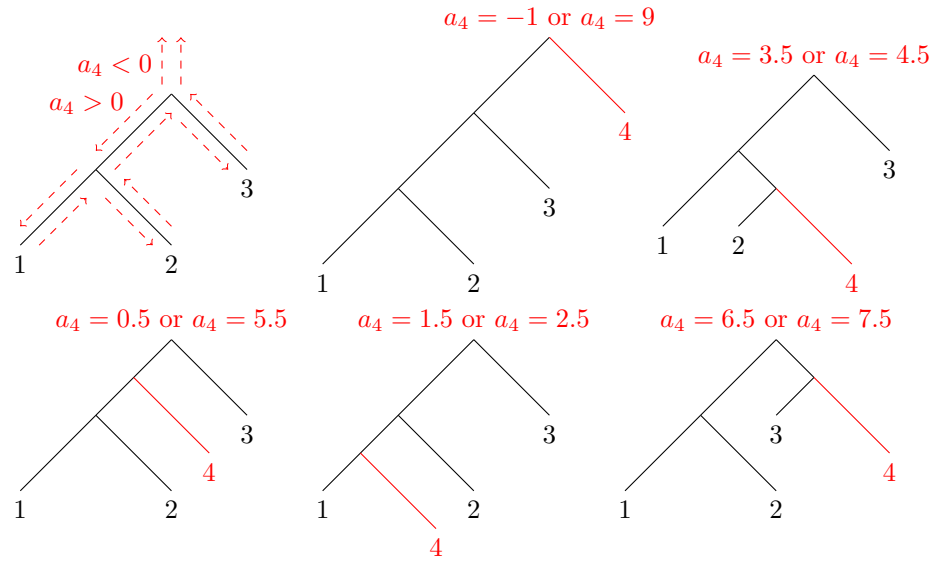


Figure 1: Illustration of the contour traversal. The initial tree is shown in black, with three leaves labeled 1, 2 and 3 and all four branches of length 1. Leaf 4 is added with the value of a_4 determining the graft point as shown and the new terminal branch of length $\exp(b_4) = 1$ shown in red.

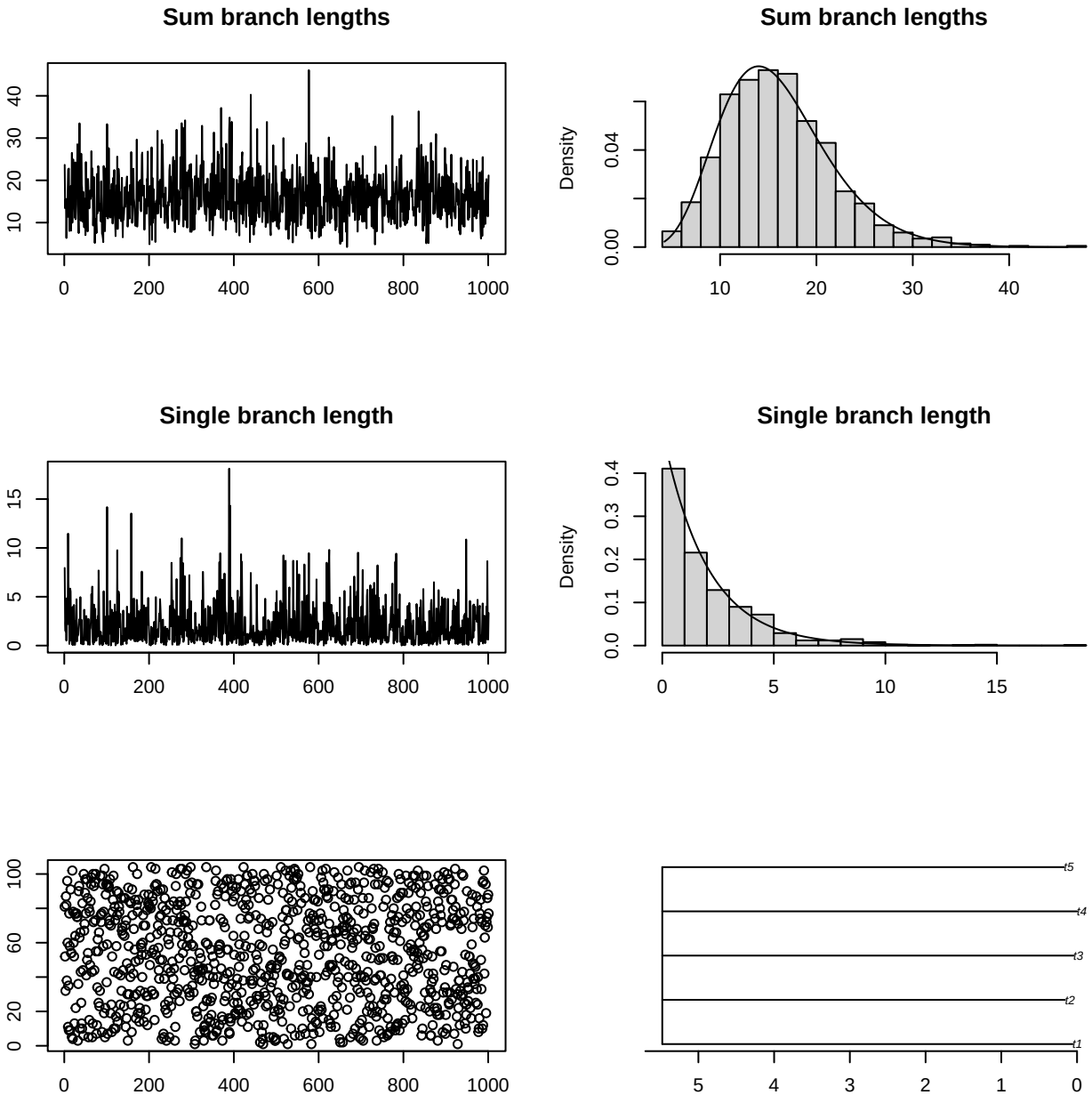


Figure 2: MCMC run. (A) Trace of sum of branch lengths. (B) Distribution of sum of branch lengths, sampled (histogram) and expected (line). (C) Trace of a single branch length. (D) Distribution of a single branch length, sampled (histogram) and expected (line). (E) Trace of topology. (F) Majority-rule consensus tree.

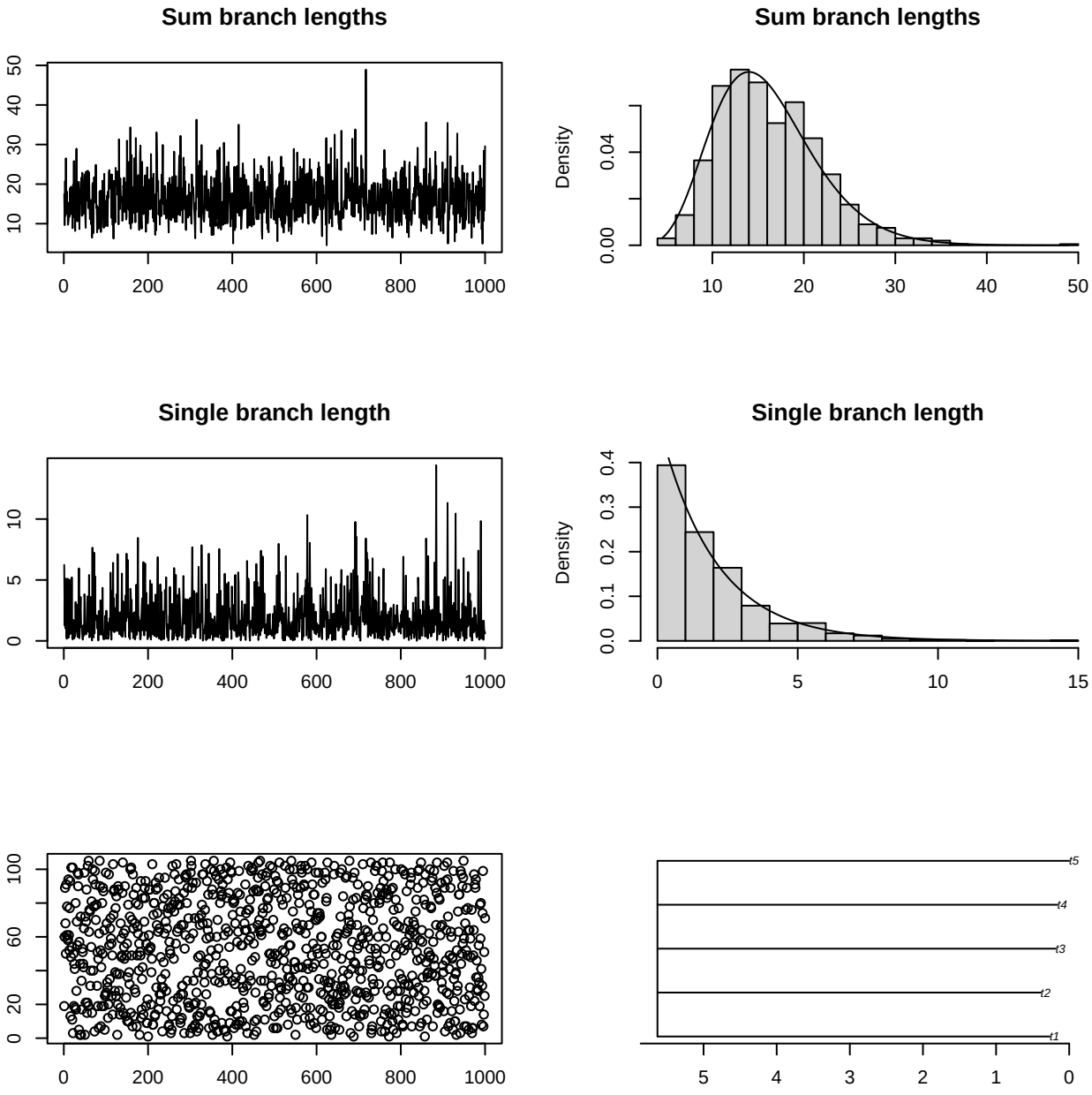


Figure 3: HMC run. (A) Trace of sum of branch lengths. (B) Distribution of sum of branch lengths, sampled (histogram) and expected (line). (C) Trace of a single branch length. (D) Distribution of a single branch length, sampled (histogram) and expected (line). (E) Trace of topology. (F) Majority-rule consensus tree.

References

Andrieu, C., Thoms, J., 2008. A tutorial on adaptive MCMC. Statistics and Computing 18, 343–373. doi:10.1007/s11222-008-9110-y.

Betancourt, M., 2017. A Conceptual Introduction to Hamiltonian Monte Carlo. arxiv arXiv:1701.02434.

Billera, L.J., Holmes, S.P., Vogtmann, K., 2001. Geometry of the Space of Phylogenetic Trees. Advances in Applied Mathematics 27, 733–767. doi:10.1006/aama.2001.0759.

Carpenter, B., Gelman, A., Hoffman, M.D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M.A., Guo, J., Li, P., Riddell, A., 2017. Stan: A probabilistic programming language. Journal of Statistical Software 76. doi:10.18637/jss.v076.i01.

Dinh, V., Bilge, A., Zhang, C., Matsen, F.A., 2017. Probabilistic Path Hamiltonian Monte Carlo. doi:10.48550/arXiv.1702.07814, arXiv:1702.07814.

Felsenstein, J., 1978. The Number of Evolutionary Trees. Systematic Zoology 27, 27. doi:10.2307/2412810.

Felsenstein, J., 1981. Evolutionary trees from DNA sequences: A maximum likelihood approach. Journal of Molecular Evolution 17, 368–376. doi:10.1007/BF01734359.

Gilks, W.R., Richardson, S., Spiegelhalter, D.J., 1996. Markov Chain Monte Carlo in Practice. volume 39. Chapman & Hall/CRC. doi:10.2307/1271145.

Höhna, S., Drummond, A.J., 2012. Guided Tree Topology Proposals for Bayesian Phylogenetic Inference. Systematic Biology 61, 1–11. doi:10.1093/sysbio/syr074.

Jukes, T., Cantor, C., 1969. Evolution of protein molecules, in: Mammalian Protein Metabolism. Academic Press, pp. 21–132.

Nascimento, F.F., Reis, M.D., Yang, Z., 2017. A biologist's guide to Bayesian phylogenetic analysis. Nature Ecology and Evolution 1, 1446–1454. doi:10.1038/s41559-017-0280-x.

Neal, R.M., 2011. MCMC Using Hamiltonian Dynamics. doi:10.1201/b10905, arXiv:1206.1901.

Paradis, E., 2006. Analysis of Phylogenetics and Evolution with R. Springer.

Plummer, M., Best, N., Cowles, K., Vines, K., 2006. CODA: Convergence diagnosis and output analysis for MCMC. R News 6, 7–11. doi:10.1159/000323281, arXiv:1011.1669v3.

Roberts, G.O., Rosenthal, J.S., 1998. Optimal Scaling of Discrete Approximations to Langevin Diffusions. Journal of the Royal Statistical Society Series B: Statistical Methodology 60, 255–268. doi:10.1111/1467-9868.00123.

Vihola, M., 2011. On the stability and ergodicity of adaptive scaling Metropolis algorithms. Stochastic Processes and their Applications 121, 2839–2860. doi:10.1016/j.spa.2011.08.006.

Warren, D.L., Geneva, A.J., Lanfear, R., 2017. RWTY (R We There Yet): An R package for examining convergence of Bayesian phylogenetic analyses. Molecular Biology and Evolution , msw279doi:10.1093/molbev/msw279.

Zhang, C., Huelsenbeck, J.P., Ronquist, F., 2020. Using Parsimony-Guided Tree Proposals to Accelerate Convergence in Bayesian Phylogenetic Inference. Systematic Biology 69, 1016–1032. doi:10.1093/sysbio/syaa002.